MIL-H-20194D
2 December 1982
SUPERSEDING
MIL-H-20194C
16 March 1977

MILITARY SPECIFICATION

HELIOTROPE, SURVEYING, BOX-TYPE, WITH CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a box-type heliotrope with carrying case.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

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MILITARY

MIL-S-3534 - Surveying Instruments and Accessories, Packaging of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13217E4800 - Heliotrope, Surveying, Box-Type, with Case.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The heliotrope and case shall be in accordance with Top Assembly TA13217E4800 and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the heliotrope. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, materials and material finishes, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.3 Materials. Materials shall be as specified herein (and on the drawings). Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Environmental conditions.

3.4.1 Operating temperature. The heliotrope shall perform as specified herein in any ambient temperature from plus 110° F (plus 43° C) to minus 25° F (minus 32° C).

3.4.2 Storage temperature. The heliotrope shall not be damaged by storage in any ambient temperature from plus 145° F (plus 63° C) to minus 28° F (minus 33° C).

3.5 Operational performance.

3.5.1 Mechanical operating conditions. All operating mechanical features shall be movable without binding of parts within the temperature extremes specified in 3.4.1.

3.5.2 Alinement. The front and rear sights shall be mounted above the alinement rings so that a line passing through the centers of the sights shall be parallel within 0.0156 inch to a line passing through the centers of the alinement rings.

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3.5.3 Mirrors. Mirrors shall be second-surface silvered on glass. The reflecting surface shall be dense, chemically deposited silver, protected by not less than a 0.001-inch-thick electrodeposited copper coating. The back and edge of each mirror shall be coated with a baked-on, weather-resistant lacquer.

3.6 Identification marking. The heliotrope and case shall be identified in accordance with MIL-STD-130.

3.7 Workmanship. All parts, components, and assemblies of the heliotrope and case shall be clean and free from corrosion, cracks, scratches, or other defects or extraneous material which will affect appearance or serviceability. All edges shall be rounded or beveled. There shall be no evidence of damaged parts or assemblies which would make the item unsatisfactory for its intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with the specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article heliotrope shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article heliotrope shall be tested as specified in 4.5.2. Failure of any test shall be cause for performing the inspection as specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.3. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The heliotropes and cases shall be examined as specified herein for the following major defects:

101. Dimensions not as specified.
102. Materials not as specified.
103. Missing components or improper assembly.
104. Mirrors not as specified.
105. Identification marking missing, incomplete, or illegible.
106. Treatment and painting not as specified.
107. Workmanship not as specified.

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4.5.2 Tests.

4.5.2.1 High temperature. The heliotrope shall be tested as specified in MIL-STD-810, procedure II, method 501.1. The maximum storage temperature shall be 145° F (63° C) and the maximum ambient operating temperature shall be 110° F (43° C). Operate the heliotrope as specified in 4.5.2.3. Nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.5.2.2 Low temperature. The heliotrope shall be tested as specified in MIL-STD-810, procedure I, method 502.1. The minimum storage temperature shall be minus 28° F (minus 33° C) and the minimum ambient operating temperature shall be minus 25° F (minus 32° C). Operate the heliotrope as specified in 4.5.2.3. Nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.5.2.3 Alinement. The sights of the front and rear vanes shall be alined by reflecting a light beam through the principal mirror to a nearby surface perpendicular to the line of sight. The shadow of the rear alinement ring shall fall on the annular band of the front ring. Evidence of the line of sight and the beam of light not parallel to within 0.0156 inch shall constitute failure of this test.

4.6 Inspection of packaging. Preservation, packing and marking for level A or level B shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534. Commercial preservation, packing, and marking shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. Each heliotrope in its case shall be preserved in accordance with the level A preservation and packaging requirements of MIL-S-3534.

5.1.2 Level B. Each heliotrope in its case shall be preserved in accordance with the level B preservation and packaging requirements of MIL-S-3534.

5.1.3 Commercial. Each heliotrope in its case shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Heliotropes preserved as specified in 5.1, shall be packed together in accordance with the level A packing requirements of MIL-S-3534.

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5.2.2 Level B. Heliotropes preserved as specified in 5.1 shall be packed together in accordance with the level B packing requirements of MIL-S-3534 except boxes conforming to PPP-B-636, V3c, style optional, may also by used. Box closure and sealing shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. Heliotropes, preserved as specified in 5.1, shall be packed together in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or level B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The heliotrope is intended for use as a target in daylight triangulation and reconnaissance surveys.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and number of units required (see 3.2, 4.3 and 6.3).

c. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a sample selected from the first production items. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

Review activities:
DLA - GS
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Preparing activity:
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